```
100 'ITERATIVE DESIGN OF RECURSIVE FILTER
110 '
120                                          'INITIALIZE
130 N%  = 256                                'number of points in FFT
140 NP% = 8                                  'number of poles in filter
150 DELTA = .00001                           'perturbation increment
160 MU = .2                                  'iteration step size
170 DIM REX[255]                             'real part of signal during FFT
180 DIM IMX[255]                             'imaginary part of signal during FFT
190 DIM T[128]                               'desired frequency response (mag only)
200 DIM A[8]                                 'the "a" recursion coefficients
210 DIM B[8]                                 'the "b" recursion coefficients
220 DIM SA[8]                                'slope for "a" coefficients
230 DIM SB[8]                                'slope for "b" coefficients
240 '
250 GOSUB XXXX                               'mythical subroutine to load T[ ]
260 '
270 FOR P% = 0 TO NP%                        'initialize coefficients to the identity system
280   A[P%] = 0
290   B[P%] = 0
300 NEXT P%
310 A[0] = 1
320 '
330 '                                        'ITERATION LOOP
340 FOR ITER% = 1 TO 100                     'loop for desired number of iterations
350   GOSUB 2000                             'calculate new coefficients
360   PRINT  ITER%  ENEW  MU                 'print current status to video screen
370   IF ENEW > EOLD THEN MU = MU/2          'adjust the value of MU
380 NEXT ITER%
390 '
400 '
410 FOR P% = 0 TO NP%                        'PRINT OUT THE COEFFICIENTS
420   PRINT  A[P%]    B[P%]
430 NEXT P%
440 '
450 END
```

TABLE 26-4

3040 to 3080 load an impulse in the array, IMX[ ]. Next, lines 3100-3150 use this impulse as an input signal to the recursive filter defined by the current values of A[ ] and B[ ]. The output of this filter is thus the *impulse response* of the current system, and is stored in the array, REX[ ]. The system's frequency response is then found by taking the FFT of the impulse response, as shown in line 3170. Subroutine 1000 is the FFT program listed in Table 12-4 in Chapter 12. This FFT subroutine returns the frequency response in rectangular form, overwriting the arrays REX[ ] and IMX[ ].

Lines 3200-3250 then calculate ER, the *mean squared error* between the magnitude of the current frequency response, and the desired frequency response. Pay particular attention to how this error is found. The iterative action of this program optimizes this error, making the way it is defined very important. The FOR-NEXT loop runs through each frequency in the frequency response. For each frequency, line 3220 calculates the magnitude of the current frequency response from the rectangular data. In line 3230, the error at this frequency is found by subtracting the desired magnitude, T[ ], from the current magnitude, MAG. This error is then squared, and added to the